legio

_travelers guide
_for navigating the legio starsystem

23	5	28
noise engineering firmwares	from the community	firmwares on one sheet

[enc] press [enc] hold › then noon = 12 o'clock cw / ccw = clockwise / counter

one 6 hp body · many minds · every control below is named by its tag on the panel drawing

labels from the noise engineering printable overlays · firmware data from each developer, 2026.10

orange = hold

01 the platform

lg_01 · fig.03

enc	encoder + cv jack (–2 to +5 v) · some firmwares add a tap
1 2	two knobs, each with a cv jack, 0 to +5 v
s1 s2	two 3-way switches, read up · mid · down
trig	trigger jack, about 1.6 v: hit, sync, fsu, mute…
in out	stereo · L feeds R when R is empty
LEDs	boot colours identify the firmware
size	6 hp · 1.5 in deep · 140 mA +12 v · 22 mA –12 v

02 swapping firmware

lg_02

1. turn off the power to your case and unscrew the legio (the usb port is on the back)
2. unplug the power cable from the module
3. plug the usb cable into the module and your computer
4. open the noise engineering firmware swap page, choose legio and a firmware, follow its steps (free, about a minute)
5. unplug usb, reconnect power, screw it back in · swap the panel overlay to match

needs a usb data cable and chrome, edge or opera · not firefox or safari

community firmware

built from source

flash	daisy patch sm in dfu mode (hold BOOT, tap RESET) · make program-dfu or dfu-util
go back	pick any noise engineering firmware on the swap page

06 community

lg_06 · continued

jawari legio

blue nautilus · github

tanpura drone: four strings strummed in turn, after the vcv module

enc tuning · push = coarse · 1 jawari shape · 2 lpf / string 1
s1 knob 2 lpf · string 1 · s2 formants on · off · trig strum

mono, same on both outs · pitch + semitone are saved

rotary legio

gugenet · github

rotary speaker: horn + drum rotors, doppler, three cabinets

enc tap = max · balance · drive · 1 doppler · 2 mic distance
s1 speed max · cv · brake · s2 cab open · classic · dark · trig max speed · in in

hold the encoder 0.8 s = bypass · enc cv jack sweeps speed

03 oscillators + voices

lg_03 · 9 firmwares

virt iter legio

manual

multi-algorithm stereo oscillator · a launch firmware

enc pitch · 1 flavor · 2 tang
s1 harm · sawx · bass s2 II · I · 0 trig sync · in pm

sinc legio

manual

stereo phase-modulation oscillator

enc pitch · 1 morph · 2 timbre
s1 warp · op · fold s2 II · I · 0 trig sync · in pm

gator legio

manual

supersaw phase-mod oscillator, stereo pm ins

enc pitch · 1 detune · 2 saw mod
s1 saw count s2 stereoness trig 8va · in pm

at legio

manual

lfsr lo-fi oscillator after the ataraxic translatron

enc pitch · 1 tone · 2 mod
s1 mode env · rm · mix s2 II · I · 0 trig sync · in in

plus legio

manual

full additive voice with envelope + lfo

enc pitch · 1 time · 2 timbre
s1 mod + · off · – s2 lfo env · on · off trig hit · in pm

03 oscillators + voices

lg_03 · continued

unda legio

manual

full phase-mod voice, built for chaining

enc pitch · 1 time · 2 timbre
s1 mod +1 · 0 · –1 s2 mode adsr · asr · ar trig hit · in mix

reso legio

manual

karplus-strong string voice + resonator

enc pitch · 1 decay · 2 brightness
s1 body banjo · violin · guitar s2 chorus II · I · 0 trig hit · in excite

marimba legio

manual

physical-model resonator voice + processor

enc pitch · 1 decay · 2 tone
s1 type tube · metal · wood s2 mode ext · bow · hit trig hit · in excite

morbus legio

manual

stereo noise generator

enc frequency · 1 harsh · in: fwr · 2 color · in: clip
s1 wtf · rough · none s2 II · I · 0 trig mute · in in

04 drums

lg_04 · 7 firmwares

kick legio

manual

analog-style kick that ducks the audio at its ins

enc pitch / mute · 1 decay · 2 impact
s1 tone s2 sat trig hit · in duck

snare legio

manual

physical-model snare

enc pitch / mute · 1 decay · 2 noise
s1 impact s2 wire trig hit · in mix

tom legio

manual

toms, lasers, zaps, mid-range hits

enc pitch / mute · 1 decay · 2 punch
s1 tone s2 sat trig hit · in mix

cymbal legio

manual

granular metal for hats, rides + crashes

enc morph / mute · 1 decay · 2 tone
s1 type ride · crash · hat s2 alloy trig hit · in mix

bos legio

manual

more cowbell, stereo pm ins

enc pitch / mute · 1 decay · 2 tone
s1 mode fm · add · x0x s2 tilt trig hit · in fm

klang legio

manual

zappy percussion after basimilus

enc pitch / mute · 1 decay · 2 impact
s1 harm s2 spread trig hit · in mix

tymp legio

manual

compact stereo digital drum

enc pitch · 1 decay · 2 wackitude
s1 mode boing · cat · boot s2 tang trig hit · in quack

06 community

lg_06 · 5 open firmwares

sawstack legio

carl tashian · github · altstrument

five-voice supersaw, no envelope · also an au / vst3 plugin · mit

enc fine pitch · push = coarse · 1 detune · 2 morph
s1 mode stack · rich · sub s2 width mono · stereo · wide trig hard sync

stutterer legio

carl tashian · github · altstrument

stutter › tape delay › dj filter › soft clip, clock-synced · mit

enc tape speed · tap = stutter · 1 subdivision · 2 dj filter
s1 clock ×2 · =1 · ÷2 s2 wet 100 · 50 · 0 % trig clock · in in

drifter legio

carl tashian · github · altstrument

random bézier auto-panner, crossfader or cv panner · mit

enc rate · tap = curve · 1 center · 2 depth
s1 mode pan · xfade · cv pan s2 edge clip · fold · wrap trig clock · in in

05 effects

lg_05 · 7 firmwares

librae legio

manual

dynamics: compress, limit, expand, gate · a launch firmware

enc color · 1 ceiling · 2 room
s1 tilt high · off · low s2 cut max · some · none trig fsu · in in

chasma legio

manual

mid / side compression + limiting

enc distortion · 1 side limit · 2 mid limit
s1 mid comp s2 side comp trig fsu · in in

ampla legio

manual

multimode filter + gate, tube-style drive

enc freq · 1 a/d · 2 time
s1 howl · bite · calm s2 hp · bp · lp trig hit · in in

roucha legio

manual

stereo multimode filter with folder

enc frequency · tap = bypass · 1 resonance · 2 fold
s1 all · post · pre s2 hp · bp · lp trig bypass · in in

bitcrush legio

manual

pitch-tracking downsampler, bits + dither

enc rate / bypass · 1 bits · 2 dither
s1 env in · out · off s2 lpf off · high · low trig bypass · in in

megafold legio

manual

stereo wavefolder + shaper with chorus

enc envelope · 1 gain · 2 shape
s1 chorus lots · some · none s2 speed fast · mid · slow trig fsu · in in

ovis legio

manual

granular shepard-tone processor

enc pitch · 1 count · 2 length
s1 fade s2 layers trig hit · in in

__fig.03 : tagged panel

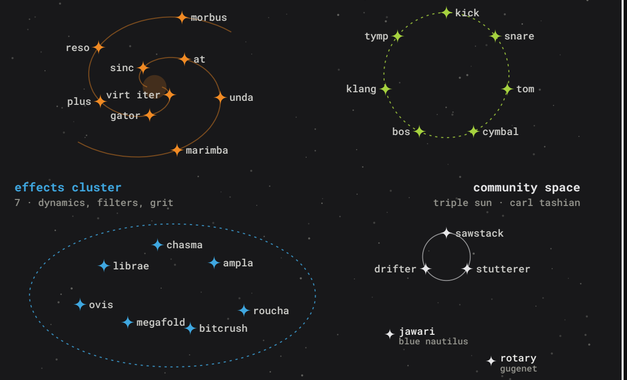
the tags every card uses



[enc]	__encoder : turn · tap in some firmwares
[1]	__top knob : the first big knob
[2]	__bottom knob : the second big knob
[s1]	__left switch : up · mid · down
[s2]	__right switch : up · mid · down
[trig]	__jack : hit / sync / fsu, about 1.6 v
[cv]	__enc · 1 · 2 : a cv jack for each
[in]	__in L / R : L feeds R if alone
[out]	__out L / R : stereo out
[LED]	__two lights : boot colours = firmware

__fig.04 : the star map

28 firmwares by family



rogue stars: one firmware per maker

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the noise engineering legio overlay template · firmware data from each developer

not affiliated with noise engineering, any firmware developer or marvel